

MAT3032 ANSWERS TO EXERCISES 2.3

Exercises 2.3

1. (a) $(i_1 \ i_2 \ \cdots \ i_k) = (i_1 \ i_k) \cdots (i_1 \ i_2)$, a product of $k - 1$ transpositions, so the number of transpositions is even if k is odd, and odd if k is even.
- (b) Suppose $\sigma = \rho_1 \cdots \rho_r$ as a product of disjoint cycles, where ρ_i is a k_i -cycle. This can be written as a product of $(k_1 - 1) + \cdots + (k_r - 1)$ transpositions. This number equals $(k_1 + \cdots + k_r) - r = n - r = n - \text{cyc}(\sigma)$, so σ is even iff $n - \text{cyc}(\sigma)$ is even.
- (c) A k -cycle $(i_1 \ i_2 \ \cdots \ i_k)$ can be expressed in cycle notation in k ways, since we can write it starting with any one of $i_1, \dots, i_k$.
- (d) An ordered list of k numbers from $\{1, \dots, n\}$ can be chosen in $\frac{n!}{(n-k)!}$ ways. Each of these represents a k -cycle but each k -cycle occurs k times by (b), so the number of k -cycles in S_n is $\frac{n!}{k(n-k)!}$.
If k is odd, a k -cycle is even so all these k -cycles are in A_n .
If k is even, a k -cycle is odd so there are none in A_n .

2. $\pi^{-1}(i'_j) = i_j$, so $\rho\pi^{-1}(i'_j) = \rho(i_j)$, which equals i_{j+1} for $j = 1, \dots, k - 1$ and equals i_1 when $j = k$.

Hence $\pi\rho\pi^{-1}(i_j) = \pi(i_{j+1})$ for $j = 1, \dots, k - 1$, and $\pi\rho\pi^{-1}(i_k) = \pi(i_1)$.

If $x \notin \{\pi(i_1), \dots, \pi(i_k)\}$ then $\pi^{-1}(x) \notin \{i_1, \dots, i_k\}$ so $\rho\pi^{-1}(x) = \pi^{-1}(x)$, hence $\pi\rho\pi^{-1}(x) = x$. It follows that $\pi\rho\pi^{-1}$ is the k -cycle $(\pi(i_1) \ \pi(i_2) \ \cdots \ \pi(i_k))$.

When $\rho = (1 \ 3 \ 6 \ 4 \ 2 \ 5)$ and $\pi = (1 \ 4 \ 6 \ 2)(3 \ 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 1 & 5 & 6 & 3 & 2 \end{pmatrix}$, we have $\pi\rho\pi^{-1} = (\pi(1) \ \pi(3) \ \pi(6) \ \pi(4) \ \pi(2) \ \pi(5)) = (4 \ 5 \ 2 \ 6 \ 1 \ 3)$.

3. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 4 & 7 & 6 & 8 \end{pmatrix} = (1 \ 2 \ 3)(4 \ 5)(6 \ 7)(8)$.
 $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 8 & 2 & 1 & 7 & 6 & 5 & 3 \end{pmatrix} = (1 \ 4)(2 \ 8 \ 3)(5 \ 7)(6)$

Both permutations have cycle type $1^1 2^2 3^1$, so they are conjugate in S_8 .

4. (These answers are not unique.)

(a) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 2 & 6 & 4 & 5 \end{pmatrix}$ or $(1)(2 \ 3)(4 \ 6 \ 5)$.

Order = $\text{lcm}(1, 2, 3) = 6$. Cycle type $1^1 2^1 3^1$, cycle index 3.

(b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 4 & 1 & 3 & 6 \end{pmatrix}$ or $(1 \ 2 \ 5 \ 3 \ 4)(6)$.

Order = 5. Cycle type $1^1 5^1$, cycle index 2.

$$(c) \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 6 & 4 & 5 & 1 & 7 \end{pmatrix} \text{ or } (1 \ 2 \ 3 \ 6)(4)(5)(7).$$

Order = 4. Cycle type $1^3 4^1$, cycle index 4.

$$5. \begin{array}{c|c|c|c|c|c|c|c|c|c|c|c|c|c|} n & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ \hline \text{Max. order} & 1 & 2 & 3 & 4 & 6 & 6 & 12 & 15 & 20 & 30 & 30 & 60 \end{array}$$

For any $n \in \mathbb{N}$, Landau's function gives the largest possible order of an element of S_n , i.e. the largest l.c.m. of any set of numbers whose sum is n . The above table gives its values for $n = 1, \dots, 12$. See Wikipedia for more details.

6. The permutations of $\{1, \dots, 10\}$ which keep each of 5, 6, 7, 8, 9, 10 fixed form a subgroup of S_{10} which is isomorphic to S_4 , so its order is 24.

If there was also a cyclic subgroup of order 24, it would be generated by an element of order 24. Writing this as a product of disjoint cycles, the *lcm* of their lengths would be 24.

Among all sets of numbers whose *lcm* is 24, the smallest sum is $3 + 8 = 11$, i.e. the permutation would have the form $(a \ b \ c)(d \ e \ f \ g \ h \ i \ j \ k)$. But $11 > 10$. Hence there is no such cyclic subgroup of S_{10} .

7. A_4 consists of the identity (1), with order 1; the 3-cycles $(1 \ 2 \ 3)$, $(1 \ 3 \ 2)$, $(1 \ 2 \ 4)$, $(1 \ 4 \ 2)$, $(1 \ 3 \ 4)$, $(1 \ 4 \ 3)$, $(2 \ 3 \ 4)$, $(2 \ 4 \ 3)$, all with order 3; and the products of disjoint transpositions $(1 \ 2)(3 \ 4)$, $(1 \ 3)(2 \ 4)$, $(1 \ 4)(2 \ 3)$, having order 2.

No element has order 6, so A_4 has no cyclic subgroup of order 6.

The three elements of order 2 in A_4 all commute with one another. This is not true of the three elements of order 2 in S_3 , e.g. $\tau(\sigma\tau) \neq (\sigma\tau)\tau$. This is a structural difference, so A_4 has no subgroup isomorphic to S_3 .

C_6 and S_3 are the only 6-element groups, so A_4 has no subgroup of order 6.

8. Suppose $\pi(1 \ 2 \ 3)\pi^{-1} = (1 \ 3 \ 2)$, so by Proposition 2.14 $(\pi(1) \ \pi(2) \ \pi(3)) = (1 \ 3 \ 2)$, which can also be written as $(2 \ 1 \ 3)$ and $(3 \ 2 \ 1)$.

π permutes $\{1, 2, 3\}$, so $\pi(4) = 4$. The possibilities are:

$\pi(1) = 1, \pi(2) = 3, \pi(3) = 2$, giving $\pi = (2 \ 3)$

or $\pi(1) = 2, \pi(2) = 1, \pi(3) = 3$, giving $\pi = (1 \ 2)$

or $\pi(1) = 3, \pi(2) = 2, \pi(3) = 1$, giving $\pi = (1 \ 3)$.

Thus π is a transposition, so $\pi \notin A_4$ and hence $(1 \ 2 \ 3)$ and $(1 \ 3 \ 2)$ are not conjugate in A_4 . (But they are, of course, conjugate in S_4 .)

9. The possible cycle types of elements in S_4 are $1^4, 1^2 2^1, 1^1 3^1, 2^2, 4^1$.

By Proposition 2.16 there are $24/24 = 1$ of type 1^4 , $24/4 = 6$ of type $1^2 2^1$, $24/3 = 8$ of type $1^1 3^1$, $24/8 = 3$ of type 2^2 and $24/4 = 6$ of type 4^1 . (Check: these add up to 24.)

The centralisers of elements with these cycle types have orders 24, 4, 3, 8, 4.

All elements of the same cycle type form a conjugacy class. Normal subgroups are unions of conjugacy classes and the order of a subgroup divides the order of the group, so the possible orders of normal subgroups of S_4 are 1, 4, 12, 24.

(These all occur. The normal subgroups of order 4 and 12 are D_2 and A_4 .)

10. (a) Elements of S_7 with cycle index 4 can have the form $(a)(b)(c)(d\ e\ f\ g)$, $(a)(b)(c\ d)(e\ f\ g)$, $(a)(b\ c)(d\ e)(f\ g)$, so the cycle types are $1^3 4^1$, $1^2 2^1 3^1$, $1^1 2^3$.

The numbers of elements of these types are $7!/4(3!)$, $7!/6(2!)$, $7!/8(3!)$, giving a total of $210 + 420 + 105 = 735$.

(b) Let σ be a p -cycle in S_p , so it has cycle type p^1 . By Proposition 2.16, the number of elements of S_p that commute with σ is p .

Now $\sigma, \sigma^2, \dots, \sigma^p$ are distinct p -cycles which clearly commute with σ (since $\sigma\sigma^k = \sigma^{k+1} = \sigma^k\sigma$ for each k), and there are p of them, so these must be precisely the elements that commute with σ .

11. $(i\ j\ k) = (i\ k)(i\ j)$, which is even, so every 3-cycle is in A_n .

$(i\ j)(i\ k) = (i\ k\ j)$ and $(i\ j)(k\ \ell) = (i\ \ell\ k)(i\ j\ k)$, so every product of pairs of transpositions is a product of 3-cycles.

Any permutation in A_n is even, i.e. a product of pairs of transpositions, so is a product of 3-cycles. (Thus the 3-cycles generate A_n .)

12. (a) $(1\ i)(1\ j)(1\ i) = (i\ j)$, so an arbitrary transposition can be expressed as a product of transpositions of the form $(1\ i)$, e.g. $(2\ 3) = (1\ 2)(1\ 3)(1\ 2)$.

S_n is generated by the set of all transpositions, hence by all those of the form $(1\ i)$ where $i \in \{2, \dots, n\}$. This generating set has $n - 1$ elements.

- (b) If $\rho = (2\ \dots\ n)$, $\tau = (1\ 2)$, we find that $\rho^r \tau \rho^{-r} = (\rho^r(1)\ \rho^r(2)) = (1\ r+2)$ for $r = 0, \dots, n-2$ so $\{\rho, \tau\}$ generates all transpositions of the form in (a), and hence all of S_n . This generating set has two elements.